WORKED SOLUTION

STEP I 2008, Question 9

Mechanics • Rolling Hoop Momentum

STEP I 2008, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q9

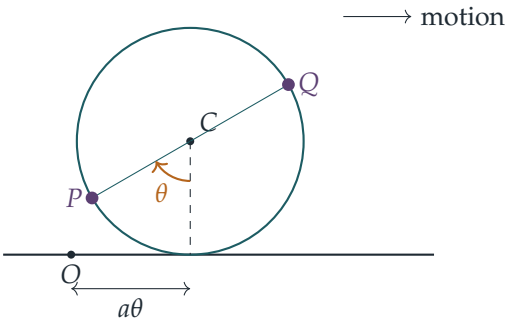
The key idea

Rolling without slipping ties the translation to the rotation: when the hoop has turned through θ, its centre has moved a distance $a\theta$, the length of rim that has touched the table. So each particle's position is "centre plus a rotating radius", and differentiating gives the velocities. For the last part, notice that the external forces have no horizontal component, so the total horizontal momentum can't change, and that momentum turns out to be a multiple of $\dot{\theta}$.

Solution

Position and speed of P

Model P and Q as particles on a light hoop. Take the origin at P's starting point, the x-axis along the line of rolling (in the direction the hoop moves) and the y-axis vertically up. Let the hoop roll in the positive x-direction, so that it turns clockwise in the picture below.



Since the hoop doesn't slip, the arc of rim that has rolled onto the table equals the distance moved by the centre, so the centre C is at $(a\theta, a)$. Initially CP points straight

down; after a clockwise turn through θ the vector $\overrightarrow{CP}$ is $(-a\sin\theta,\ -a\cos\theta)$. Hence

$$\boxed{P = (a\theta - a\sin\theta,\ a - a\cos\theta)}$$

(a cycloid). Differentiating with respect to t, using the chain rule,

$$\dot{x}_P = a\dot{\theta}(1-\cos\theta), \qquad \dot{y}_P = a\dot{\theta}\sin\theta,$$

so

$$v_P^2 = a^2\dot{\theta}^2[(1-\cos\theta)^2 + \sin^2\theta] = 2a^2\dot{\theta}^2(1-\cos\theta) = 4a^2\dot{\theta}^2\sin^2\tfrac{1}{2}\theta,$$

using $1-\cos\theta = 2\sin^2\frac{1}{2}\theta$. The speed of P is therefore

$$v_P = 2a\,|\dot{\theta}\sin\tfrac{1}{2}\theta| \qquad (= 2a\dot{\theta}\sin\tfrac{1}{2}\theta \text{ for } 0 \le \theta \le 2\pi,\ \dot{\theta} \ge 0).$$

This vanishes when θ is a multiple of 2π, exactly when P is the point of contact, as rolling requires.

Kinetic energy

Q is at the other end of the diameter, so $\overrightarrow{CQ} = -\overrightarrow{CP}$ and

$$Q = (a\theta + a\sin\theta,\ a + a\cos\theta), \qquad \dot{x}_Q = a\dot{\theta}(1+\cos\theta), \quad \dot{y}_Q = -a\dot{\theta}\sin\theta.$$

In the same way,

$$v_Q^2 = a^2\dot{\theta}^2[(1+\cos\theta)^2 + \sin^2\theta] = 2a^2\dot{\theta}^2(1+\cos\theta) = 4a^2\dot{\theta}^2\cos^2\tfrac{1}{2}\theta.$$

The hoop is light, so the kinetic energy is that of the particles alone:

$$T = \tfrac{1}{2}mv_P^2 + \tfrac{1}{2}mv_Q^2 = 2ma^2\dot{\theta}^2(\sin^2\tfrac{1}{2}\theta + \cos^2\tfrac{1}{2}\theta) = 2ma^2\dot{\theta}^2,$$

as required.

Constant speed

The external forces are the weights mg of the particles and the reaction of the table, and all of them are vertical. So the total horizontal force on the system is zero, and Newton's second law for the system says its total horizontal momentum is constant. (The hoop is light, so it carries no momentum.)

Ask yourself Why is the horizontal momentum enough, when the particles also move vertically?

Because the horizontal momentum happens to depend only on $\dot{\theta}$. Adding the horizontal velocities,

$$m\dot{x}_P + m\dot{x}_Q = ma\dot{\theta}(1-\cos\theta) + ma\dot{\theta}(1+\cos\theta) = 2ma\dot{\theta}.$$

The $\cos\theta$ terms cancel: the system's centre of mass is C, which always moves horizontally with velocity $a\dot{\theta}$. Since $2ma\dot{\theta}$ is constant, $\dot{\theta}$ is constant, and the hoop's speed (the speed $a\dot{\theta}$ of its centre) is constant. □

Or by energy. The heights of P and Q add to $(a - a\cos\theta) + (a + a\cos\theta) = 2a$, so the potential energy $2mga$ never changes. The reaction acts at the point of contact, which is instantaneously at rest when rolling, so it does no work. Hence $T = 2ma^2\dot{\theta}^2$ is constant, so $|\dot{\theta}|$ is constant; and since $\dot{\theta}$ varies continuously and can't jump from one sign to the other without passing through 0, $\dot{\theta}$ itself is constant. The momentum argument avoids this last step, which is why we used it first.

Checks

- At $\theta = \pi$, P is at the top: the formula gives $P = (a\pi,\, 2a)$ and speed $2a\dot{\theta}$, twice the speed of the centre, the familiar result for the top of a rolling wheel.
- The kinetic energy equals $\frac{1}{2}(2m)(a\dot{\theta})^2 + \frac{1}{2}(2ma^2)\dot{\theta}^2$: translation of the centre of mass plus rotation about it, with moment of inertia $2ma^2$. Both terms give $ma^2\dot{\theta}^2$.
- Numerically, with $a = 1.3$, $\dot{\theta} = 2.1$ and $\theta = 1$, differentiating the positions by finite differences gives $v_P^2 + v_Q^2 = 4a^2\dot{\theta}^2 \approx 29.8$, and the horizontal accelerations of P and Q cancel when $\dot{\theta}$ is constant.

Common slips

- Putting the centre at $(a,\, a)$ or forgetting it moves at all: the $a\theta$ term is what rolling contributes.
- Getting the sign of $\sin\theta$ wrong in P's position; check that P starts with zero velocity.
- Concluding constant speed from constant kinetic energy without saying why $\dot{\theta}$ can't change sign.

Beyond STEP

This is a two-particle example of the general fact that the total momentum of a system changes only through external forces: internal forces (here carried by the hoop) cancel in pairs. The kinetic-energy split in the Checks is König's theorem, kinetic energy equals that of the centre of mass plus that of the motion relative to it, which is the starting point for rigid-body dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rolling motion and these ideas carefully.

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